



- Note:
- Attempt all questions
 - Assume suitable data if necessary
 - Use of non-programmable calculator is permitted.
 - This paper contains 4 pages.

Q.N.		Question	Marks	CO	BT
Q.1		MCQ (0.5 × 20 = 10 Marks)			
	1	Which of the following best explains why edges are relatively invariant to global illumination changes? A) Edges are computed from absolute intensity values. B) Edges depend on differences in local intensity rather than global values. C) Edge detection involves histogram equalization. D) Edge maps are unaffected by lighting in all cases.	0.5	CS416.2	BT3
	2	In the Canny edge detector, the role of non-maximum suppression is to: A) Reduce noise by smoothing the image B) Suppress weak edges based on a threshold C) Thin edges to one-pixel width by retaining only local maxima D) Remove edges caused by illumination changes	0.5	CS416.2	BT2
	3	Why is Gaussian smoothing applied before edge detection? A) To increase edge sharpness B) To suppress high-frequency noise that could cause false edges C) To detect corners instead of edges D) To normalize intensity values	0.5	CS416.2	BT2
	4	Which of the following situations will cause false edges in an image? A) Global brightness increase B) Addition of salt-and-pepper noise C) Uniform contrast stretching D) Histogram equalization	0.5	CS416.2	BT3
	5	The Laplacian of Gaussian (LoG) detects edges by: A) Computing the first derivative of intensity values B) Finding zero-crossings in the second derivative of intensity C) Enhancing high-frequency Fourier components D) Applying thresholding on the gradient magnitude	0.5	CS416.2	BT4
	6	Which parameter in the Canny edge detector controls the trade-off between detecting true edges and avoiding noise sensitivity ? A) Gaussian kernel size (σ) B) Non-maximum suppression window size C) Double threshold ratio D) Edge linking connectivity	0.5	CS416.2	BT4
	7	Which statement about edge invariance is incorrect ? A) Edges are robust to uniform contrast changes B) Edges are fully invariant to scaling C) Edges are relatively stable under mild illumination changes D) Edges may disappear under strong blurring	0.5	CS416.2	BT3

8	In the Canny edge detector, if the lower threshold is set too high relative to the upper threshold: A) Only very strong edges will be detected, and edge continuity may break B) Weak edges will always be linked to strong edges C) False edges due to noise will increase D) It has no effect since only the upper threshold matters	0.5	CS416.2	B
9	A LoG (Laplacian of Gaussian) operator with a large Gaussian kernel size will: A) Detect more fine edges with higher accuracy B) Suppress fine details and detect only coarse/strong edges C) Increase false edges due to noise D) Behave like Roberts operator	0.5	CS416.2	BT5
10	In frequency domain, applying a high-pass filter corresponds to: A) Enhancing global illumination B) Detecting edges by retaining sharp intensity transitions C) Blurring the image to remove noise D) Histogram equalization	0.5	CS416.2	BT4
11	Which of the following is a drawback of zero-crossing edge detectors (like Marr-Hildreth)? A) They cannot detect diagonal edges B) They are not rotationally invariant C) They produce double edges around intensity changes D) They cannot be implemented in the frequency domain	0.5	CS416.2	BT2
12	Which of the following best explains why hysteresis thresholding is superior to single-threshold methods in Canny? A) It eliminates strong edges entirely B) It links weak edges to strong ones, reducing fragmentation C) It increases sensitivity to noise D) It avoids Gaussian smoothing	0.5	CS416.2	BT5
13	In edge detection, which trade-off is described by the "uncertainty principle of scale"? A) High localization accuracy vs. high detection accuracy B) Fine edge detection vs. robustness to noise C) Gradient magnitude vs. direction accuracy D) Computational efficiency vs. edge strength	0.5	CS416.5	BT5
14	In Harris corner detection, a "corner" is identified when: A) One eigenvalue of M is large, the other small B) Both eigenvalues of M are small C) Both eigenvalues of M are large D) Eigenvalues of M are equal	0.5	CS416.5	BT4
15	The Harris response function is given by: $R = \det(M) - k \cdot (\text{trace}(M))^2$ What is the role of the empirical constant k? A) It balances sensitivity between corners and edges B) It adjusts Gaussian smoothing scale C) It removes noise in gradient computation D) It determines the size of the sliding window	0.5	CS416.5	BT4
16	If in Harris corner detection, $\lambda_1 \gg \lambda_2 \approx 0$, what is the likely interpretation of the pixel? A) Flat region B) Edge C) Corner D) Noise point	0.5	CS416.5	BT5

17	Which scenario causes Harris corner detection to fail without modifications? A) Image rotation B) Global illumination changes C) Large scale changes D) Additive Gaussian noise	0.5	CS416.5	BT3
18	What effect does increasing the Gaussian window size in Harris detection have? A) Detects finer corners with better localization B) Smooths out noise but may miss fine corners C) Eliminates edges completely D) Increases eigenvalues without changing detection	0.5	CS416.5	BT5
19	Why does Harris use $\det(M)$ and $\text{trace}(M)$ instead of directly computing eigenvalues? A) They are faster to compute and equivalent up to constants B) Eigenvalues require Fourier transform C) Eigenvalues cannot be computed for singular matrices D) Eigenvalues are unstable under rotation	0.5	CS416.1	BT4
20	Which of the following is true regarding the rotation invariance of Harris detector? A) It is not rotation-invariant because gradients depend on direction B) It is rotation-invariant because eigenvalues of M are unchanged under rotation C) It is rotation-invariant only for circular windows D) It requires histogram equalization for rotation invariance	0.5	CS416.5	BT4
Q.2	True / False Questions (0.5 × 10 = 5 Marks)			
1	Gradient-based edge detectors provide both edge strength and orientation.	0.5	CS416.2	BT2
2	The Laplacian operator detects edges by computing the magnitude of the first derivative.	0.5	CS416.2	BT3
3	Edge detection in images can be formulated as a problem of detecting discontinuities in intensity.	0.5	CS416.5	BT3
4	Edge maps generated from second derivative operators are less likely to produce double edges than first derivative operators.	0.5	CS416.5	BT4
5	Anisotropic diffusion is sometimes used as a preprocessing step to preserve edges while reducing noise.	0.5	CS416.1	BT2
6	In practice, edge linking algorithms are necessary to form continuous contours from broken edge fragments.	0.5	CS416.5	BT2
7	The Marr-Hildreth detector is essentially a zero-crossing detector applied after Gaussian smoothing.	0.5	CS416.2	BT3
8	Harris corners are detected by directly computing the eigenvalues of the structure tensor matrix.	0.5	CS416.5	BT2
9	In Harris corner detection, a flat region corresponds to both eigenvalues of the structure tensor being close to zero.	0.5	CS416.5	BT4
10	Increasing the window size in corner detection always improves detection of small corners.	0.5	CS416.5	BT4

Q.3	Fill in the blank questions (0.5 × 10 = 5 Marks)					
1	In a 3×3 image gradient patch, the computed gradients are $I_x=2$, $I_y=3$. The structure tensor at this point is $M = \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$. The value of $\det(M)$ = _____.	0.5	CS416.5	BT1		
2	A Harris corner detector computes eigenvalues $\lambda_1 = 0.8$, $\lambda_2 = 0.7$ at a pixel. Using $R = \det(M) - k \cdot (\text{trace}(M))^2$ with $k = 0.05$. The value of R is _____.	0.5	CS416.5	BT5		
3	For a 3×3 Sobel kernel applied to a pixel neighborhood with intensity values $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$, the gradient in the X direction G_x is _____.	0.5	CS416.2	BT5		
4	Using the same matrix as Q3, the gradient in the Y direction G_y is _____.	0.5	CS416.2	BT5		
5	In a Harris corner detector, if the structure tensor has eigenvalues $\lambda_1 = 0.2$, $\lambda_2 = 0.02$, and $k = 0.05$, the response function R = _____.	0.5	CS416.5	BT5		
6	A LoG kernel is generated with $\sigma = 1$. If the kernel size is calculated as $6\sigma + 1$, the kernel size is _____.	0.5	CS416.2	BT3		
7	In a 5×5 Prewitt kernel, the sum of all kernel elements is _____.	0.5	CS416.2	BT5		
8	In a Harris corner detector, the structure tensor at a pixel is $M = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$ with $k=0.04$. The Harris response $R = \det(M) - k \cdot (\text{trace}(M))^2$ is _____.	0.5	CS416.5	BT5		
9	A 2×2 image patch has pixel intensities: $\begin{bmatrix} 100 & 150 \\ 120 & 130 \end{bmatrix}$. The robert cross operator uses kernels: $G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. The gradient magnitude $\sqrt{G_x^2 + G_y^2}$ is _____.	0.5	CS416.2	BT5		
10	A Canny edge detector is applied to an image with the following parameters: Gaussian smoothing $\sigma = 1.0$, Upper threshold for hysteresis = 120, Lower threshold = 40% of the upper threshold. The lower threshold value is _____. If a pixel has a gradient magnitude of 50, will it be classified as a strong edge, weak edge, or non-edge? = _____.	0.5	CS416.2	BT4		

Course Outcomes (CO)

CS416.1 - Describe different image representations, their mathematical representation and data structures used.

CS416.2 - Classify different Boundary Detection algorithms.

CS416.3 - Create a 3D object from a given set of images.

CS416.4 - Detect a moving object in video using the concept of motion analysis.

CS416.5 - Recognize the object using the concept of computer vision.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



**Shri Guru Gobind Singhji Institute of Engineering
and Technology, Nanded-431606**

ISE-II Exam

Subject: **Distributed Computing**

Date: 10/10/2025

Time: 14.30 To 15.15

Code: **PCC-CS404**

Max. Marks: 20

Instructions:

1. All questions are compulsory.
2. Assume suitable data if necessary.

Q.1 Write in brief to solve the Following (Any Two)

10

- a. Explain the steps in creating a cluster using Hadoop.
- b. Explain processing challenges of Big Data and options to its processing approaches.
- c. Explain attribute based naming used in Distributed System.
- d. Describe how connection-less communication between a client and a server proceeds when using sockets.

Q.2 a. Explain different approaches used to resolve Flat Names in distributed System. 05

- b. Describe actions to be taken with references to local resources when migrating code to another machine. 05

End



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of COMPUTER SCIENCE
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Cryptography and
Network Security

Course Code: PCC-CS-403

Date: 11-10-2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Proper steps indicate full marks.**
- The symbols carry usual meaning.

Q.N.	Solve any 4 Questions	Marks	CO	BT
Q.1	Find the multiplicative inverse of 23 modulo 120 using the Extended Euclidean Algorithm.	05	CO1	BT1 BT3
Q.2	Solve the system of congruences using Chinese Remainder Theorem : $x \equiv 2 \pmod{3}$ $x \equiv 3 \pmod{4}$ $x \equiv 2 \pmod{5}$	05	CO1	BT2 BT3
Q.3	Use Fermat's algorithm to factorize $N=2491$.	05	CO1	BT1 BT3
Q.4	Find whether 17 and 3120 are co-prime?	05	CO1	BT4
Q.5	Use the Miller-Rabin test to determine if $n=221$ is composite or prime? Take base $a=2$	05	CO1	BT3 BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the number theory concepts
CO 2	Impart knowledge on symmetric and asymmetric encryption techniques.
CO 3	Create Authentication functions as Message Authentication Codes and Hash Functions
CO 4	Examine the issues and structure of Authentication Service and Electronic Mail, web and IP security.
CO 5	Provide familiarity in system security.



**SGGS Institute of Engineering and Technology,
Vishnupuri, Nanded**

In Semester Evaluation- II (Semester-I) 2025

Final Year B. Tech.

Computer Science & Engineering

TCP/IP Networking

PCC-CS402

Time: 14:30 to 15:15

Max. Marks : 20

Date:

13/10/2025

- Note:
- i. Attempt all questions
 - ii. Assume suitable data if necessary
 - iii. Use of non-programmable calculator is permitted.
 - iv. Figures to the right indicate Marks

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|-----|--|----|
| Q.1 | Construct a routing table by giving suitable example. | 06 |
| Q.2 | Explain ARP packet format. | 05 |
| Q.3 | What are the three ranges of port numbers used in UDP. | 03 |
| Q.4 | Demonstrate IP Checksum calculation with suitable example. | 06 |